# Zongyuan Li

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## Research Interests

Regularity theory of elliptic and parabolic PDEs.

## Employment

**City University of Hong Kong**, Assistant Professor (Tenure-Track) 2024 –
**Rutgers University**, Hill Assistant Professor (Non Tenure Track) 2020 – 2023

## Education

**Brown University**, Ph.D. in Applied Mathematics 2015 – 2020

Thesis: Elliptic boundary value problems on irregular domains.
Advisor: Prof. Hongjie Dong.

**Fudan University**, B.Sc. in Mathematics 2010 – 2015

Thesis: Birkhoff normal forms in some PDEs.
Advisor: Prof. Xiaoping Yuan.

**University of Oxford**, Part C in Math 2013 – 2014

Disseratation: Study of Keldysh Equation and Related PDEs
Thesis supervisor: Prof. Gui-Qiang G. Chen and Dr. Wei Xiang.
College Tutor (St. Hugh's College): Dr. Tom Sanders.

## Long-term Visits

American Institute of Mathematics, Pasadena, CA, USA Apr 2024, May 2025, Aug 2026
Institute for Advanced Study, Princeton, NJ, USA Jun 2024
SUNY Binghamton, Vestal, NY, USA Aug – Dec 2023

## Grants, Honors & Awards

### Grants

RGC General Research Fund (GRF) # 9044066 (HKD 616,160). 2027 – 2029
NSFC Young Scientists Fund, #12501275 (RMB 300,000). 2026 – 2028
RGC Early Career Scheme (ECS), # 21307225 (HKD 868,287). 2025 – 2028
CityU Start-up Grant, # 7200810 (internal, HKD 300,000). 2024 – 2026

AMS–Simons Travel Grants (USD 5000). 2021 – 2023

**Honors & awards**

Sigma Xi Award, Brown Applied Math. 2020
Outstanding Graduate (ranked 1st in math major), Fudan University. 2015
Bronze Medals in Probability and Statistics, S.-T. Yau College Student Math Contests. 2013, 14
Bronze Medal in team competition, S.-T. Yau College Student Math Contests. 2013
First Prize, Chinese Mathematics Competition (Final Round at Tongji University). 2012
First Prize, China National High School Math League. 2009

## Publications

**Papers**

1 Liouville Theorem with Boundary Conditions from Chern-Gauss-Bonnet Formula. (With B.Z. Chu and Y.Y. Li) Accepted by **Amer. J. Math.** arXiv:2410.16384.
2 Asymptotic expansions for harmonic functions at conical boundary points. (With D. Kriventsov) **Rev. Mat. Iberoam.**, 41(3): 837–866, 2025.
3 Unique continuation for Robin problems with non-smooth potentials. **J. Funct. Anal.**, 288 (6):Paper No. 110811, 2025.
4 On the uniqueness of variable coefficient Schrödinger equations. (With S. Federico and X. Yu) **Commun. Contemp. Math.**, 27 (3), 2025.
5 Mixed boundary value problems for parabolic equations in Sobolev spaces with mixed-norms. (With J. Choi and H. Dong) **Calc. Var. Partial Differential Equations.**, 62(1): Paper No. 5, 2023.
6 Quantitative unique continuation for Robin boundary value problems on $C^{1,1}$ domains. (With W. Wang) **Indiana Univ. Math. J.**, 72(4):1429–1460, 2023.
7 The Dirichlet-conormal problem for the heat equation with inhomogeneous boundary conditions. (With H. Dong) **Adv. Math.**, 411: Paper No. 108777, 2022.
8 Optimal regularity of mixed Dirichlet-conormal boundary value problems for parabolic operators. (With J. Choi and H. Dong) **SIAM J. Math. Anal.**, 54(2):1393-1427, 2022.
9 On the $W^2_p$ estimate for oblique derivative problem in Lipschitz domains. (With H. Dong) **Int. Math. Res. Not. IMRN**, (5):3602-3635, 2022.
10 Norm inflation for the Boussinesq system. (With W. Wang) **Discrete Contin. Dyn. Syst. Ser. B**, 26(10):5449-5463, 2021.
11 The conormal and Robin boundary value problems in nonsmooth domains satisfying a measure condition. (With H. Dong) **J. Funct. Anal.**, 281(9):Paper No. 109167, 32, 2021.
12 The Dirichlet-conormal problem with homogeneous and inhomogeneous boundary conditions. (With H. Dong) **Comm. Partial Differential Equations**, 46(3):470–497, 2021.
13 Optimal regularity for a Dirichlet-conormal problem in Reifenberg flat domain. (With J. Choi and H. Dong) **Appl. Math. Optim.**, 83(3):1547-1583, 2021.
14 Classical solutions of oblique derivative problem in nonsmooth domains with mean Dini coefficients. (With H. Dong) **Trans. Amer. Math. Soc.**, 373(7) : 4975-4997, 2020.

**Book Chapters**

1 Asymptotics of harmonic functions in the absence of monotonicity formulas. In *Extended Abstracts 2021/2022*, Trends in Mathematics, vol 3, pages 125 – 131. Birkhäuser, 2024.

**Preprints**

1 Liouville theorems for conformally invariant fully nonlinear equations. I (With B. Z. Chu and Y. Y. Li) arXiv:2311.07542.
2 On the fully nonlinear Yamabe problem with constant boundary mean curvature. I (With B. Z. Chu and Y. Y. Li) arXiv:2410.09683.
3 On one-space dimensional parabolic equations with measurable coefficients: Sobolev estimates and the Alexandrov maximum principle. (With H. Dong) arXiv:2609.25568.
4 On the $W^{2,p}$ solvability for mixed boundary value problems. (With R. Khandelwal) arXiv:2609.30090

## Mentorship

**Ph.D. student**

| | |
|---|---|
| Yuhong XU | 2026 – |

**Others**

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| Sirui ZHANG and Chun Hing LAM, undergraduate final year project at CityUHK | 2026 – 27 |
| Yuhong XU, undergraduate final year project at CityUHK | 2025 – 26 |
| Cheng Shing CHUA, summer research internship student from NUS | 2025 |
| Ruixiu LYU, Ms.C. (Financial Mathematics and Statistics) dissertation at CityUHK | 2024 – 25 |

## Teaching

**CityUHK Math**

| | |
|---|---|
| MA 4545 (Undergraduate Differential Geometry) | 2025, 2026 Fall |
| MA 2185 (Direcrete Math for Computer Sciences) | 2025, 2026 Fall |
| MA 3512 (Undergraduate PDE) | 2024 Fall |
| MA 3526 (Analysis) | 2024, 2025, 2026, 2027 Spring |

**Rutgers Math**

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|---|---|
| Math 421 (Calculus for Engineer) | 2022 Spring |
| Math 252 (ODE) | 2021 Spring, 2022 Fall, 2023 Spring |
| Math 423 (Undergraduate PDE) | 2020, 2021, 2022 Fall |

**Brown Applied Math (TA)**

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|---|---|
| AM 2640 (Graduate Probability II) | 2020 Spring |
| AM 2550 (Graduate Numerical PDE I) | 2019 Fall |
| AM 2110 (Graduate Real Analysis I) | 2018 Fall |
| AM 33, 34 (ODE I, II) | 2016 Fall, 2017 Spring |

## Talks

Analysis seminar, Southern University of Science and Technology. Nov. 2026
PDE conference, Westlake University. Sep. 2026
Applied math seminar, Oklahoma State University. Aug. 2026
Analysis seminar, University of Oklahoma. Aug. 2026
PDE seminar, Westlake University. Jun. 2026
Winter workshop on geometric PDEs, Central South University. Dec. 2025
PDE seminar, Lanzhou University. Dec. 2025
PDE seminar, Duke Kunshan University. Oct. 2025
Symposium on elliptic and parabolic PDEs at Yanqi Lake, Beijing. Oct. 2025
Short course at Beijing Normal University. Aug. 2025
Conformal and CR geometry and their geometric flows, Sanya, China. Jul. 2025
Recent Advances in Potential Theory and Partial Differential Equations, OIST, Japan. Jul. 2025
International Workshop on Fully Nonlinear Partial Differential Equations, Fudan University. Jun. 2025
First Math Workshop of CityU and the University of Edinburgh , Hong Kong. May 2025
Joint meeting of the NZMS, AustMS and AMS, New Zealand. Dec. 2024
Modern Methods in Nonlinear Elliptic and Parabolic PDE, Bulgaria. May 2024
International Conference on Elliptic and Parabolic Problems: GAETA 2024, Italy. May 2024
PDE seminar, Hong Kong Polytechnic University. Mar. and Apr. 2024
AMS sectional meeting, Creighton University. Oct. 2023
PDE seminar, Brown University (online). Sep. 2023
Analysis seminar, Temple University. Sep. 2023
Analysis seminar, SUNY Binghamton. Sep. 2023
Colloquium, University of Toronto Scarbourough (online). Apr. 2023
Colloquium, City University of Hong Kong (online). Apr. 2023
AMS sectional meeting, University of Cincinnati. Apr. 2023
Colloquium, University of Nebraska Lincoln. Dec. 2022
PDE seminar, Brown University. Dec. 2022
Ghent Methusalem Junior Analysis and PDE Seminar, Ghent University (online). Nov. 2022
Prairie Analysis Seminar, University of Kansas. Oct. 2022
AMS sectional meeting, University of Tennessee at Chattanooga. Oct. 2022
AMS sectional meeting, University of Texas at El Paso. Sep. 2022
Analysis seminar, University of Arizona (online). Mar. 2022
International Conference on Partial Differential Equations Related to Material Science, Beijing (online). May 2021
Learning Seminar on PDE and Applications, Rutgers University (online). Mar. and Apr. 2021
Virtual Analysis and PDE Seminar (VAPS), hosted by Purdue University (online). Jan. 2021

New postdoc special colloquium, Rutgers University (online). Oct. 2020
Learning Seminar on PDE and Applications, Rutgers University (online). Aug. 2020
AMS sectional meeting, Tufts University (canceled due to COVID-19). Mar. 2020
Brown-BU-UMassAmherst PDE and Dynamics seminar. Nov. 2019
PDE seminar, Brown University. Nov. 2019
2019 Northeast Analysis Network Conference, University of Connecticut. Sep. 2019
PDE seminar, Brown University. Oct. 2018

## Services

### Conferences organized

Special session *Recent Developments in Elliptic PDEs and Related Problems*. JMM 2027, Chicago, USA Jan 2027 (Scheduled)

International conference on partial differential equations, Hong Kong Jan 2027 (Scheduled)

Special session *Recent Developments in Regularity Theory for PDEs*, The 15th AIMS conference, Athens, Greek Jul 2026

Special session *Recent Advances in Elliptic PDEs and Their Applications*. JMM 2026, Washington, DC., USA Jan 2026

Special session *Recent Developments in PDEs and Related Areas*, JMM 2025, Seattle, USA Jan 2025

Special session *New trends in elliptic and parabolic PDEs*, The 14th AIMS Conference, Abu Dhabi, UAE Dec 2024

Special session *Dynamics and Regularity of PDEs*, JMM 2024, San Francisco, USA Jan 2024

Special session *Topics in PDEs and harmonic analysis*, AMS Fall Eastern Sectional Meeting, UMass Amherst, USA Oct 2022

Special session *Regularity Theory for Linear and Nonlinear PDEs*, AMS Spring Western Sectional Meeting (online, formerly at San Francisco State University, USA) May 2021

**Reviewed journal articles** (quick opinion and full report)

*Adv. Math.*, *Adv. Nonlinear Anal.*, *Ann. Appl. Math.*, *Appl. Math. Lett.*, *Commun. Anal. Geom.*, *Complex Var. Elliptic Equ.*, *Calc. Var. Partial Differential Equations*, *Commun. Pure Appl. Anal.*, *Discrete Contin. Dyn. Syst. Ser. B*, *J. Differential Equations*, *J. Math. Pures Appl.*, *Int. Math. Res. Not. IMRN*, *NoDEA Nonlinear Differential Equations Appl.*, *Nonlinearity*, *Nonlinear Anal.*, *Trans. Amer. Math. Soc.*, *Math. Z*